Name ____ Date ___________ Period _____

Writing the Inverse of a Linear Function - Practice

Algebra II Honors | Unit 1 | Lesson 11 Practice | TEK A2.2.B

Objective: I can use the general inverse formula for a linear function, and reason about self-inverse lines and fixed points.

Section 1 — Recall

1. Using $f^{-1}(x) = (x - b)/m$, find the inverse of $f(x) = 7x + 3$ by plugging directly into the formula ($m = 7$, $b = 3$).

$f^{-1}(x)$ = ______________________

2. For $f(x) = mx + b$ with $m = -4$, what is the slope of f^{-1}?

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Section 2 — Apply

3. Derive the inverse of $g(x) = (2x - 9)/5$, showing the swap-and-solve steps.

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$g^{-1}(x)$ = ________________________

4. Find the inverse of $h(x) = (x + 12)/(-3)$, then verify by checking that $h(h^{-1}(0)) = 0$.

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$h^{-1}(x)$ = ___________________

Section 3 — Challenge

5. A self-inverse line (other than $y = x$) has the form $f(x) = -x + b$. Prove that $f(f(x)) = x$ for every x, for any value of b.

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$f(f(x))$ = ___________________________________

6. The Celsius/Fahrenheit conversion $F(c) = (9/5)c + 32$ and its inverse meet at $c = -40$. Find where the Celsius-to-Kelvin conversion $K(c) = c + 273$ meets its own inverse (solve $K(c) = c$ for c).

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Section 4 — Explain It

7. A classmate claims that because $f(x) = mx + b$ and its inverse always have reciprocal slopes, and 1 is its own reciprocal, $f(x) = x$ must be the only self-inverse line with positive slope. Is the classmate correct? Explore whether any other positive slope could work.

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